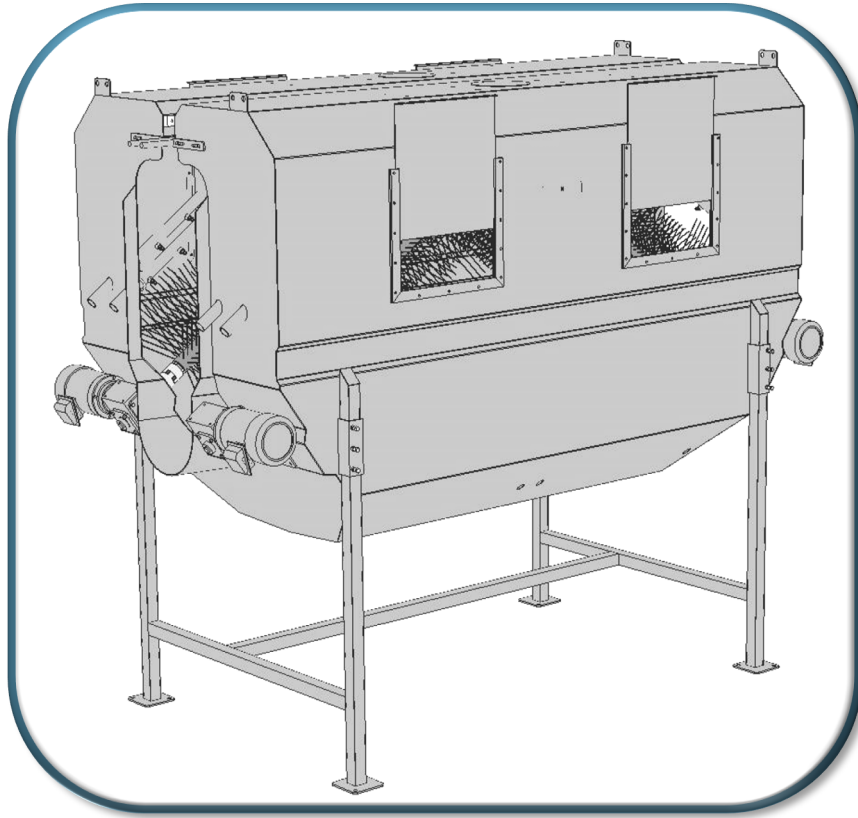


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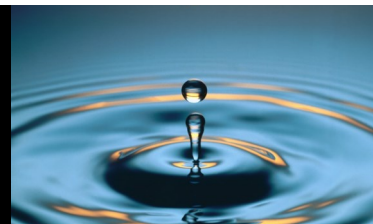
EVIS ULTIMATE ONE BIRD WASHER
UPDATED 22 JUNE 2021

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Safety

Overview

Operation

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Parts

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Notes

This Equipment is patented. The material in this manual is proprietary. This manual has been prepared for the use of the purchaser and their agents.

Reproduction and or distribution outside the purchaser's control is strictly prohibited.

Safety

**FF-2016EWUO
Ultimate One Bird Washer**

SAFETY WARNINGS

DANGER - Serious injury may occur if physical contact is made. This system contains moving and rotating parts that must be avoided.

DO NOT service or clean while the system is in operation

DO NOT place or extend any objects into the system while it is in operational mode.

CAUTION - System may start without warning - Do Not enter unless in lockout!

CAUTION - Multiple electrical components are used to operate this system. Follow all established safety procedures when servicing or cleaning this system.

All operating controls should be placed so as to be outside the normal operating area of the system.

After servicing or cleaning, remove all service items, ladders, tools, hoses etc., before operating system.

Follow all established lock out/tag out procedures.

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OVERVIEW

This system has two sections, the bottom to top trailing edge wash and the top to bottom leading edge wash. Both sections are based on the same proven principle of replacing large volumes of high pressure water with the mechanical action of rotating multi-length bristle brushes and low pressure low volume water.

Overview

The entrance end of the cabinet houses the bottom to top trailing edge wash. This wash consists of two banks of brushes, one bank on each side of the bird with the brushes rotating down. There are also two sets of two water headers, one set on each side of the bird. The outer headers clean the brushes. The other header is positioned over and parallel to the brush. Its purpose is to facilitate the contaminate removal process focusing on the trailing side of the bird.

The exit end of the cabinet houses the top to bottom leading edge wash. This wash consists of two banks of brushes, one bank on each side of the bird with the brushes rotating down on the breast and the back. There are also two sets of two water headers, one set on each side of the bird. The outer headers clean the brushes. The other header is positioned over and parallel to the brush. Its purpose is to facilitate the contaminate removal process focusing on the leading side of the bird.

The brushes are constructed of Nylon 6 Plus[®] multi-length bristles on PVC cores with stainless steel set collars with stainless steel set screws. They come in 12" lengths and 21" and 24" diameters. The 21" and 24" diameter brushes are placed across and next to each other to move the bird back and forth so that the brushes have better access to the wings. They rotate at 175 RPM.

The continued success of this system is dependent upon a careful adherence to recommended operating parameters and procedures as well as a good Preventive Maintenance program.

Some details in the maintenance section are common procedures and have been omitted.

The headers are populated with Set & Forget nozzles that are always in the correct alignment. See the maintenance section for additional information.

Questions should be addressed to Water Management Resources at
800-552-5797 ~ 678-794-7123 ~ btg@watermr.com

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OPERATION

Electrical: 4 - 2.0 HP 480 VAC 5.70/2.85 amps each

Drains: 1 - 4" or 6" Dia. w/Sanitary Ferrule

Water: 41 GPM @ 50 PSI and 50 GPM @ 75 PSI
8 - 1.25" headers connections

Header and Nozzle Orientation

Product Headers - Will be properly aligned during installation and welded in place.

Nozzles - The *Set & Forget Nozzle* housing will be installed and properly aligned during installation and Loctite® will be added to secure it in place. NOTE: The nozzles will not be installed until the lines have been flushed.

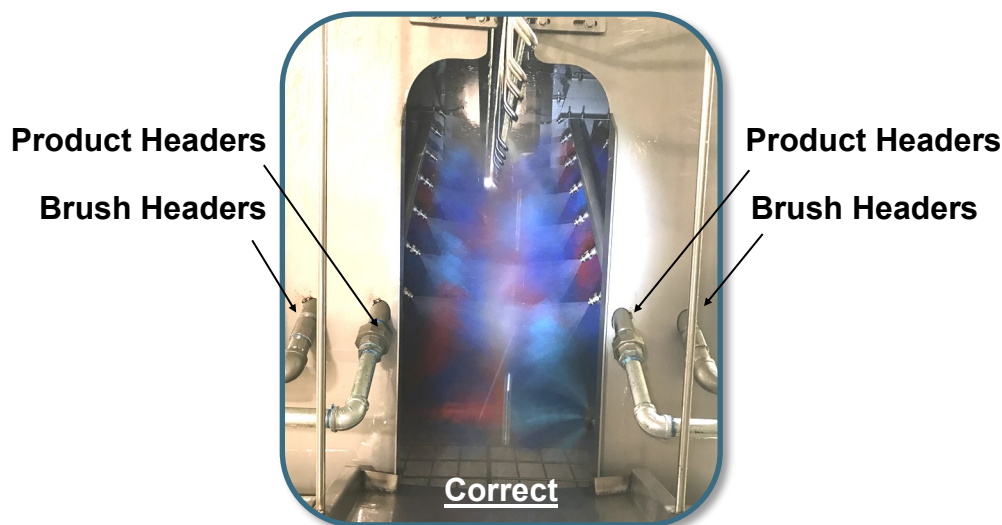
Brush Headers - Will be properly aligned during installation and welded in place.

Nozzles - The *Set & Forget Nozzle* housing will be installed and properly aligned during installation and Loctite® will be added to secure it in place. NOTE: The nozzles will not be installed until the lines have been flushed.

Here is a link to a video of the correct spray pattern.

<https://youtu.be/qnl6WUgM0oE>

SPRAY PATTERN



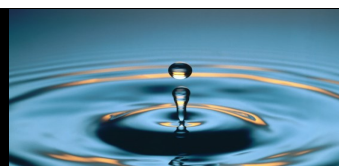
Operation

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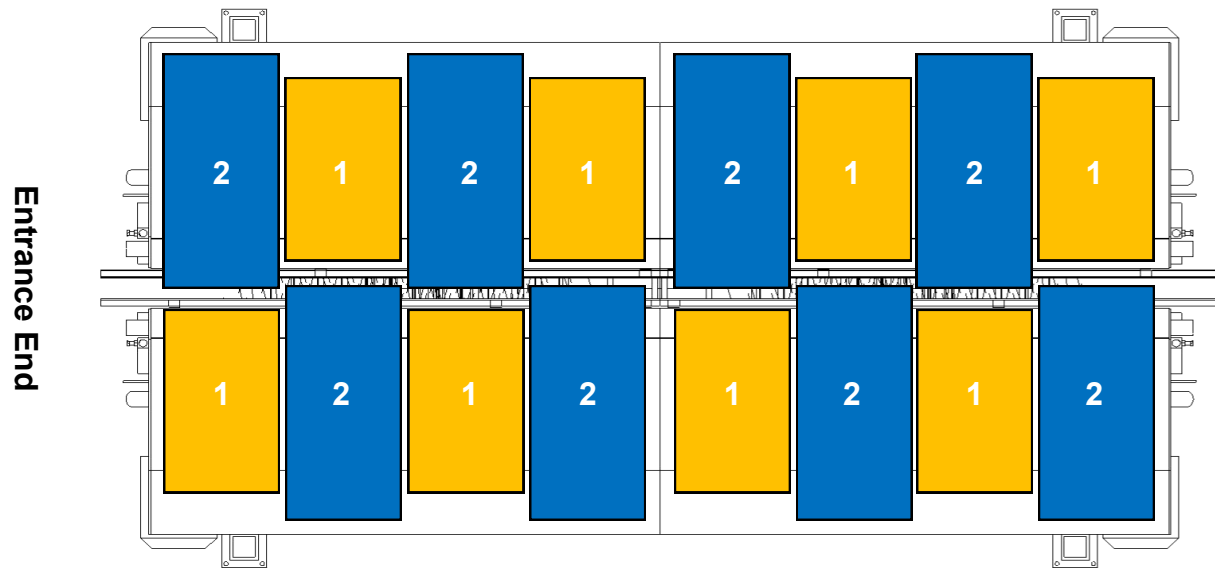
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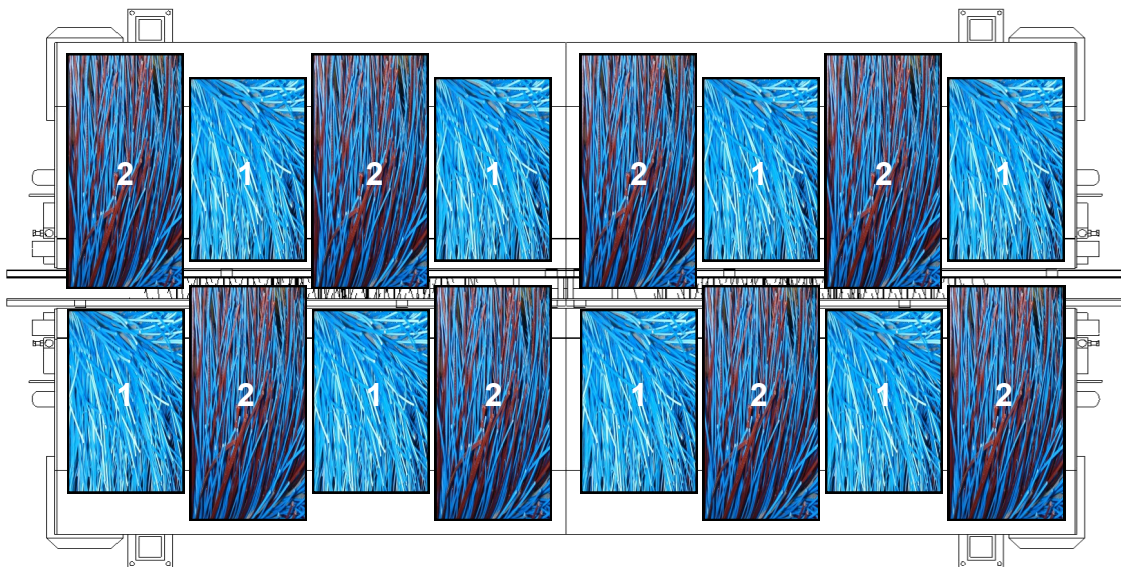


Brush Placement



Brushes

1 - 12"L x 21"Dia. 2 - 12"L x 24"Dia.



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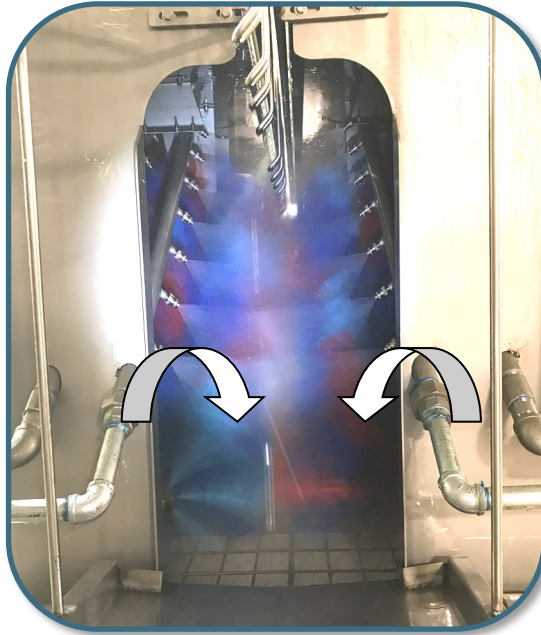
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Brush Rotation

Entrance End

Breast



Back

Operation

Exit End

Back



Breast

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Preventative Maintenance

The following chart indicates suggested inspection intervals

Item	Daily	Weekly	Monthly	Notes
Bearings		X		Inspect for wear
Motors	X			Inspect during operation for over heating
	X			Inspect during operation for Bearing Failure
Brushes		X		Inspect for breakage and wear
Coupling			X	Check for cracks and metal fatigue
Coupling Bushing		X		Check for excessive wear
Nozzles	X			Check for proper alignment of spray pattern
	X			Check for blockage or obstruction of fluid flow
			X	Inspect the orifice for wear which has changed the spray pattern and flow rate
Guide Bars	X			Check for proper spacing 1.5 x the dia. of shackle

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Extending Brush Life

A 4 to 6 week PM schedule of rotating the brushes may extend the brush life and could increase it's cleaning effectiveness. There are specific ways to achieve the rotating task without removing the brushes from the shaft.

Entrance End—Remove both sets of brushes (shaft and brushes) and move each set to the other side. See the maintenance section for shaft removal procedures.

Exit End—Remove both sets of brushes (shaft and brushes) and move each set to the other side. See the maintenance section for shaft removal procedures.



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Maintenance - Nozzle Configuration/Cleaning/Replacement

Configuration

Brush Header	All	# Needed	Alignment
Breast Side	8003	14	Parallel
Back Side	8003	14	Parallel
Product Header	All	# Needed	Alignment
Breast Side	5010	14	Vertical
Back Side	5010	14	Vertical

Nozzle Cleaning/Replacement:

Remove the nozzle by turning it 1/4 turn counter-clockwise ☹. You should be able to do this procedure with your hand. Use caution when removing and replacing the nozzle with a tool.

Replace the nozzle by aligning the nozzle threads with the Set & Forget Nozzle Housing threads and turning 1/4 turn clockwise ☺.

NOTES:

Product Headers - Align the headers so that the center of the spray pattern is aimed at the tip of the brushes. The **Nozzle** spray pattern is vertical.

Brush Headers - Align the headers so that the center of the spray pattern is aimed is at the back half of the brushes nearest the cabinet wall. The **Nozzle** spray pattern is running parallel with the brushes.

Maintenance

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Maintenance TORQUE GUIDE

BOLT SIZE	TYPE 304 Stainless Steel.	TYPE 316 Stainless Steel
2-56	2.5	2.6
2-64	3.0	3.2
3-48	3.9	4.0
3-56	4.4	4.6
4-40	5.2	5.5
4-48	6.6	6.9
5-40	7.7	8.1
5-44	9.4	9.8
6-32	9.6	10.1
6-40	12.1	12.7
8-32	19.8	20.7
8-36	22.0	23.0
10-24	22.8	23.8
10-32	31.7	33.1
¼"-20	75.2	78.8
¼"-28	94.0	99.0
⅜"-18	132	138
⅜"-24	142	147
½"-16	236	247
½"-24	259	271
⅝"-14	376	393
⅝"-20	400	418
½"-13	517	542
½"-20	541	565
⅝"-12	682	713
⅝"-18	752	787
¾"-11	1110	1160
¾"-18	1244	1301
¾"-10	1530	1582
¾"-16	1490	1558
⅞"-9	2328	2430
⅞"-14	2318	2420
1"-8	3440	3595
1"-14	3110	3250
1⅛"-7	413	432
1⅛"-12	390	408
1¼"-7	523	546
1¼"-12	480	504
1½"-6	888	930
1½"-12	703	732

Source: ITT Harper

Suggested Max Torquing Values—a guide based upon industry tests on dry products wiped clean. Values thru 1" diameter are stated in inch pounds; over 1" diameter, in foot pounds. The ⅜" diameter and under metal products were roll-threaded and, where size range permitted, were made on Bolt Maker equipment.

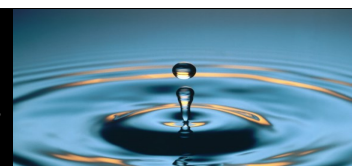
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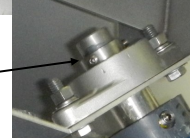
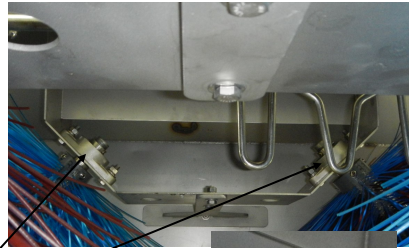
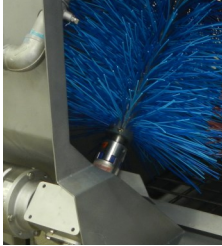
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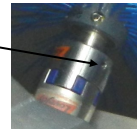


Maintenance

Coupling Replacement



1. Initiate established lock out/tag out procedures.
2. Locate and loosen the set screws on the bearing and lock collar.
3. Remove the lock collar and set aside.
4. Slide the brush shaft up and away from the coupling.
It may be necessary to move the brushes down on the shaft to make space for the shaft to slide up and out of the coupling. To do this, loosen the set screws on each end of all the brushes.
5. Locate and loosen the set screws on each half of the coupling.
6. Remove lock collar and the coupling.
7. Remove the bushing and lay aside.
8. Place one half of the new coupling on the motor jack shaft and the other half on the brush shaft.
9. Tighten the set screws on both halves of the coupling.
10. Inspect the bushing and if serviceable, seat it in the coupling. If the bushing is not serviceable, replace with a new one.
11. Insert the brush shaft into the bearing. Slide the brush shaft down and rotate to mate the two halves of the coupling.
12. Replace the lock collar on the bearing end of the brush shaft.
13. Tighten the set screws on the bearing and lock collar.
14. Recheck your work to insure all set screws are secure.
15. Check the area to insure all tools and equipment have been removed.
16. Secure the area and follow established lock out/tag out removal procedures.



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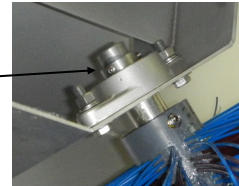


Maintenance

Brush Replacement



1. Initiate established lock out/tag out procedures.
2. Locate and loosen the set screws on the bearing and lock collar.
3. Remove the lock collar and set aside.
4. Slide the brush shaft up and away from the coupling.
It may be necessary to move the brushes down on the shaft to make space for the shaft to slide up and out of the coupling. To do this, loosen the set screws on each end of all the brushes.
5. Remove the brush shaft from the cabinet.
6. Remove the defective or worn brush(es) from the shaft.
Before removing the brushes, note the location of each type of brush. Misplacement of brush types will effect the system in properly cleaning the wings.
7. To install a new brushes and tighten all the brush set screws.
8. Insert the brush shaft into the bearing. Slide the brush shaft down and rotate to mate the two halves of the coupling.
9. Replace the lock collar on the bearing end of the brush shaft.
10. Tighten the set screws on the bearing and lock collar.
11. Recheck your work to insure all nuts are secure.
12. Check the area to insure all tools and equipment have been removed.
13. Secure the area and follow established lock out/tag out removal procedures.



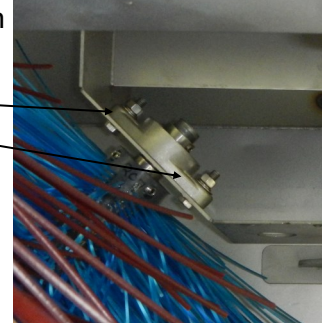
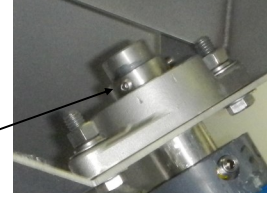
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Maintenance

Bearing Replacement

1. Initiate established lock out/tag out procedures.
2. Locate and loosen the set screws on the locking collar and remove the collar.
3. Locate and loosen the set screws on the bearing then loosen and remove the two bolts on the bearing.
4. Slide the bearing off the end of the shaft.
5. Slide the replacement bearing onto the shaft.
6. Replace the bolts and tighten the nuts.
7. Replace the lock collar on the shaft.
8. Tighten the set screws on the bearing and lock collar.
9. Recheck your work and rotate the brush shaft to insure ease of rotation.
10. Check for tools you used to insure they have not been left in the work area.
11. Secure the area and follow established lock out/tag out removal procedures.



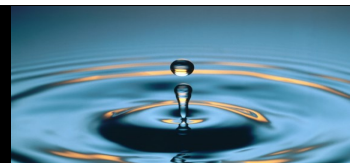
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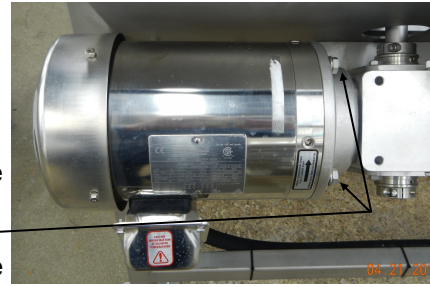
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Maintenance

Motor Replacement

1. Initiate established lock out/tag out procedures.
2. With the electrical off, disconnect the incoming power line from the junction box on the motor.
3. Locate and loosen the bolts securing the motor to the gearbox.
4. Remove the defective motor.
5. Replace the new motor aligning the motor holes with the gearbox.
6. Insert the bolts through the motor and secure.
7. Recheck your work to insure all set screws and nuts are secure.
8. Check the area to insure all tools and equipment have been removed.
9. Secure the area and follow established lock out/tag out removal procedures.
10. After the system is powered up, check for fluid leaks.



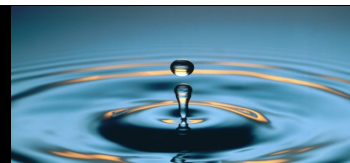
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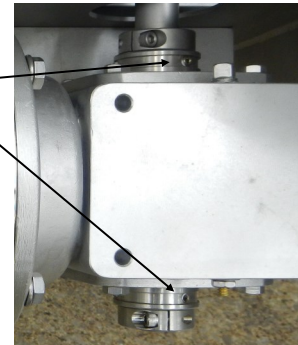
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Maintenance

Gearbox Replacement

1. Initiate established lock out/tag out procedures.
2. With the electrical off, disconnect the incoming power line from the junction box on the motor.
3. Secure or remove the brush shaft.
4. Locate and loosen the bolts securing the motor to the gearbox and remove the motor
5. Remove the lock collar from the end of the jack shaft.
6. Locate and loosen the bolts securing the gearbox to the motor mount and remove the gearbox.
7. Loosen the set screws securing the gearbox to the jack shaft and remove the jack shaft.
8. Install the new gearbox, motor, jack shaft and coupling reversing the removal steps.
9. Align gearbox, coupling, and bearing IAW appropriate tolerances as suggested by coupling manufacturer. Shimming of motor/bearing may be necessary and is considered normal maintenance practice. Misalignment may reduce bushing and coupling life span and could increase down time.
10. Recheck your work to insure all set screws and nuts are secure.
11. Check the area to insure all tools and equipment have been removed.
12. Secure the area and follow established lock out/tag out removal procedures.



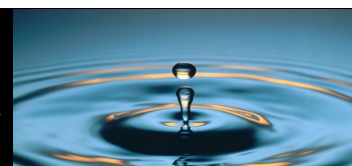
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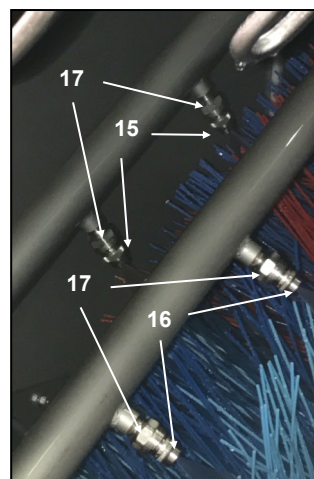
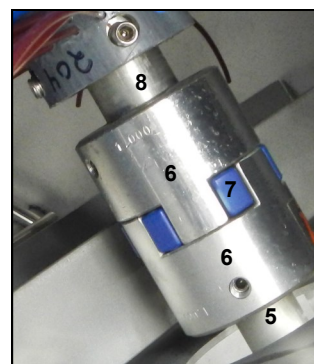
Replacement Parts

FF-2016EWUO Washer

Part Category *

	Part Number	Description	A	B
1	212-20S	Sterling Stainless Steel 175 RPM Gearbox (2206 Reducer)	X	
2	212-31S-F1	Sterling Stainless Steel 2.0 HP Motor Left hand Mount	X	
3	212-31S-F2	Sterling Stainless Steel 2.0 HP Motor Right Hand Mount	X	
4	212-10	Bearing 1"	X	
5	216-40JS	Jack Shaft for FF-2016EWUO	X	
6	212-40LJC-L100	Stainless Steel L100 Lovejoy coupling w/Heavy Duty spider bushing	X	
7	212-40LJCB-L100	Spider bushing heavy duty for L100 LJC	X	
8	216-40MSE	Main Shaft for FF-4016EUO	X	
9	413-60-12ML21090MC	Brush for FF-4016UO / FF-2016EWUO 12"L x 21" Dia.		X
10	413-60-12ML24090MC	Brush for FF-4016UO / FF-2016EWUO 12"L x 24 Dia.		X
11	Not used			X
12	Not used			X
13	413-95	Booster Pump for FF-4016UO 1.5 HP (OPTIONAL Not Shown)		X
14	Not used			X
15	N8003* Set & Forget Nozzle	Stainless Steel 8003 Spray Set & Forget Nozzle		X
16	N5010* Set & Forget Nozzle	Stainless Steel 5010 Spray Set & Forget Nozzle		X
17	Set & Forget Nozzle Housing	*Call for information		X

* Part Category: A - System Down B - System Up but less effective



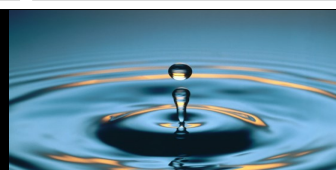
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Parts

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Sanitation

1. Observe all company safety rules and procedures.
2. Always follow company lock out and tag out procedures.
3. Perform dry pickup of all loose debris in the area and on equipment.
4. Rinse down the systems, starting at the top and working down, using plant water – preferably hot water if available.
5. Inspect to make sure all debris is rinsed off. If not, re-rinse or remove any debris left by hand.
6. Apply standard chemical cleaning solution used in the plant to the equipment. This equipment is made to withstand cleaning using alkaline foaming cleaners commonly used in the plant sanitation process.
7. Rinse thoroughly.
8. Apply final sanitizer.
9. Follow-up lockout and tag out procedures prior to start-up.

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Sanitation



CONTACTS

Replacement Parts Order

Office 706-612-7483
FAX 803-426-1392
Email sales@watermr.com

Field Sales

Denson Green

Office 910-850-5146
Email dg@watermr.com

Greg Bilyeu

Office 620-431-6303
Email gb@watermr.com

Engineering / General Information

Braiden Griffiths

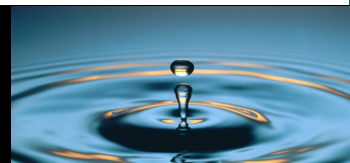
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